

Ex.no:3	DATA CLEANING

Aim:

To clean and preprocess the Diabetes Prediction Dataset by identifying and handling missing values, removing duplicate and blank records, standardizing categorical values, correcting data types, and preparing a consistent and reliable dataset for further diabetes analysis.

Algorithm:

Check Missing Values

1. Start.
2. Import the Pandas library.
3. Load the Diabetes Prediction dataset into a DataFrame (df).
4. Display dataset information using df.info().
5. Check for missing values using df.isnull().
6. Count missing values in each column using df.isnull().sum().
7. Check missing values using df.isna().
8. Check non-missing values using df.notnull().
9. Display the missing-value results.

Drop Missing Values

1. Start.
2. Import the Pandas library.
3. Load the Diabetes Prediction dataset into a DataFrame (df).
4. Drop rows containing missing values using dropna().
5. Use axis=0 to remove rows or axis=1 to remove columns.
6. Use how='any' to remove rows or columns containing any missing value.
7. Use how='all' to remove rows or columns only when all values are missing.
8. Use thresh=N to retain rows or columns with at least N non-missing values.
9. Use subset=['column_name'] to remove rows based on specific columns.
10. Use inplace=True to modify the original DataFrame.
11. Display the cleaned Diabetes Prediction dataset.

Program:

```
import pandas as pd

df = pd.read_csv("diabetes_prediction_dataset.csv")

print("\nOriginal Dataset:")
display(df.head())

print("\nDataset Information:")
df.info()

print("\nMissing Values using isnull():")
display(df.isnull())

print("\nMissing Values Count:")
print(df.isnull().sum())

print("\nMissing Values using isna():")
display(df.isna())

print("\nMissing Values Count using isna():")
print(df.isna().sum())

print("\nNon-Missing Values:")
display(df.notnull())

print("\nTotal Missing Values:")
print(df.isnull().sum().sum())

df = df.dropna(how="all")
```

```
df = df.drop_duplicates()

df["gender"] = df["gender"].astype(str).str.strip().str.title()

df["smoking_history"] = (
    df["smoking_history"]
    .astype(str)
    .str.strip()
    .str.lower()
)

df["age"] = pd.to_numeric(df["age"], errors="coerce")
df["hypertension"] = pd.to_numeric(
    df["hypertension"], errors="coerce"
)
df["heart_disease"] = pd.to_numeric(
    df["heart_disease"], errors="coerce"
)
df["bmi"] = pd.to_numeric(
    df["bmi"], errors="coerce"
)
df["HbA1c_level"] = pd.to_numeric(
    df["HbA1c_level"], errors="coerce"
)
df["blood_glucose_level"] = pd.to_numeric(
    df["blood_glucose_level"], errors="coerce"
)
df["diabetes"] = pd.to_numeric(
    df["diabetes"], errors="coerce"
)
```

```
df = df.dropna()
```

```
print("\nMissing Values After Cleaning:")
```

```
print(df.isnull().sum())
```

```
print("\nDuplicate Records After Cleaning:")
```

```
print(df.duplicated().sum())
```

```
print("\nCleaned Dataset:")
```

```
display(df.head())
```

```
print("\nCleaned Dataset Information:")
```

```
df.info()
```

```
print("\nCleaned Dataset Shape:")
```

```
print(df.shape)
```

```
df.to_csv(  
    "cleaned_diabetes_dataset.csv",  
    index=False  
)
```

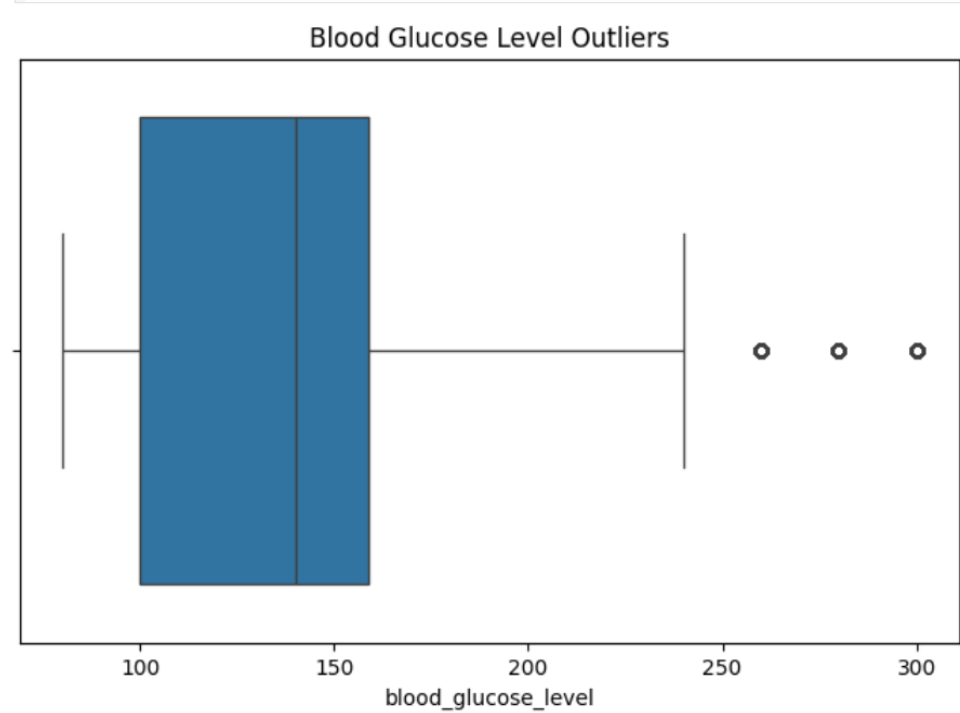
```
print("\nCleaned dataset saved successfully.")
```

Output:

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
0	Female	80.0	0	1	never	25.19	6.6	140	0
1	Female	54.0	0	0	No Info	27.32	6.6	80	0
2	Male	28.0	0	0	never	27.32	5.7	158	0
3	Female	36.0	0	0	current	23.45	5.0	155	0
4	Male	76.0	1	1	current	20.14	4.8	155	0

	gender	age	hypertension	heart_disease	smoking_history	bmi	\
38	Male	50.0	1	0	current	27.32	
94	Female	60.0	0	0	never	27.32	
104	Female	80.0	1	0	never	27.32	
125	Female	77.0	0	0	never	31.70	
142	Female	80.0	0	0	former	21.97	

	HbA1c_level	blood_glucose_level	diabetes
38	5.7	260	1
94	7.5	300	1
104	6.8	280	1
125	6.5	280	1
142	7.0	300	1



Result: